Python Engineering - Class 03 Theory Guide

Loops and Iteration

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. Iteration and recurrence

A loop is appropriate when the next action depends on repeating the same transformation across indices, terms or time steps.

Relationship / equation

```
state_{k+1} = F(state_k)
```

Connection to code

Accumulators and time-stepping methods are both recurrence relations.

2. Accumulators

A sum or product can be built from a neutral initial value.

Relationship / equation

```
sum starts at 0; product starts at 1
```

Connection to code

Choosing the wrong initial value changes the entire result.

3. Taylor approximation

Taylor series replace a smooth function by a polynomial around an expansion point. Truncating after N terms creates approximation error.

Relationship / equation

```
exp(x)=1+x+x^2/2!+...
```

Connection to code

The loop constructs one term at a time; N controls computational effort and truncation.

4. Explicit Euler

Euler replaces continuous derivatives by finite time-step updates based on the slope at the current state.

Relationship / equation

```
y_{k+1}=y_k+f(t_k,y_k)*dt
```

Connection to code

For vertical motion, the state is [h,v]. Smaller dt usually improves accuracy but increases iterations.

5. Analytical vs numerical validation

A numerical answer should be checked against a known solution whenever one exists.

Relationship / equation

```
h(t)=h0+v0*t-0.5*g*t^2
```

Connection to code

EX08 compares maximum altitude, time of flight and impact velocity to analytical formulas.

Manual validation method

For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

Loops are the basis of iterative numerical methods, time integration, Monte Carlo and grid searches.

Final summary

```
MODEL / CONCEPT
      |
      v
ALGORITHM
      |
      v
PYTHON IMPLEMENTATION
      |
      v
REFERENCE CHECK
      |
      v
ENGINEERING INTERPRETATION
```

> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.